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Generative AI chatbot for Student Counselling

Udit Kumar Reg No: RA2011031010057
Priyanshu Maheshwari Reg No: RA2011028010091
Batch ID: NWC150057
Guide name and Designation: Dr. A. Helen Victoria (Assistant Professor)

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Introduction

Mental well-being issues are a widespread global problem. About 29% of people worldwide experience them at some point in their lives, with anxiety and depression being the most common. These conditions not only reduce quality of life but also lead to substantial economic burdens, including indirect costs like reduced workplace productivity. This underscores the need for improved global mental health resources.

Literature Review

PAPER TITLE	OBJECTIVE	METHODOLOGY	ADVANTAGES	LIMITATIONS/ FUTURE SCOPE
A Mental Health Chatbot for Regulating Emotions (SERMO) - Concept and Usability Test 1 July 2021 IEEE	Address the global shortage of human resources delivering mental health services by introducing a mobile application called SERMO	SERMO to provide CBT-based emotional support via NLP and daily interactions.	Personalized, continuous monitoring, education, and self-help for global mental health support	Ilexicon-based approach for emotion detection in NLP may not accurately capture complex or subtle emotions
AI-Based Deep Learning Chatbot for Career and Personal Mentorship 16 June 2023 IEEE	virtual mentor chatbot for career guidance and emotional support	Creating a dataset for training and utilizing TensorFlow and Keras for machine learning models. HTML and CSS are used for the graphical user interface, while Flask integrates the system into a user-friendly web application	This system serves as a convenient and accessible alternative to traditional mentoring, providing students with tailored support	While it offers anonymity and convenience, it may lack the personalized, empathetic touch of in-person mentoring sessions.
Machine Learning based Intelligent Career Counselling Chatbot (ICCC). 24 May 2023	Intelligent Career Counselling Chatbot (ICCC) capable of providing career guidance to 10th and 12th-grade students. Additionally, the chatbot aims to assist students in making informed	user-centric design, data selection from YML files, and the utilization of the Chatterbot module for conversational interaction. Machine learning algorithms, including	The utilization of the Random Forest algorithm increases prediction reliability by aggregating	While machine learning algorithms improve accuracy, they can also introduce complexity,

Career Counseling Chatbot using Microsoft Bot Frameworks1 11 May 2022 IEEE	Aims to support students in identifying and planning their future career paths based on their profiles	Microsoft technologies, including QnA , Bot Composer for integration	QnA Maker allows easy creation of a knowledge base for tailored responses.	Licensing and operational costs for Azure services may be involved, depending on usage and scale.
Mind Relaxation Chatbot for University Students by Using Dense Neural Network 5 January 2022 IEEE	Aims to address this issue by creating a chatbot that can provide a non-judgmental and supportive platform for students to express their thoughts and worries openly	Data collection, preprocessing, DNN model for chatbot development using NLP	Give suggestions to students to overcome stress during academic pressure and deadline on projects, helping them to manage user experience.	Ethical considerations arise when using chatbots in sensitive areas like mental healthcare.
AI-Driven Bilingual Talkbot for Academic Counselling 24 July 2023 IEEE	Develop a multilingual Talbot for efficient, domain-specific, and responsive user support.	deep learning, NLP, and domain-specific knowledge to create a user-friendly, multilingual interface.	The use of a domain-specific knowledge base allows for tailored responses to specific educational queries.	The use of translation APIs for language conversion, especially for Arabic, may introduce errors or inconsistencies in the responses.
Dost-Mental Health Assistant Chatbot 13 February 2023 IEEE	To provide accessible mental health support and guidance through empathetic and informative conversations.	Employs Rasa's NLU pipeline, leveraging AI, natural language processing, and emotion recognition to understand and respond	The chatbot offers easily accessible mental health support to a wide range of users via the Telegram application, the chatbot's ability to use predefined stories and rules allows for a tailored and flexible user experience.	The chatbot may have limitations in addressing complex mental health issues, as it primarily relies on predefined responses and rules

Existing Challenges

From a research point of view, the challenges faced in existing counselling chatbots include:

1. **Generalization**: Getting a general answer may not be apt for every user, instead it should be modifiable with respect to user prompt rather than just matching the words in data and giving answer.

2. **Maintaining History**: Not able to maintain a record or history of the previous conversation, which could lead to wrong/irrelevant results for a user using that chatbot.

3. **Hallucination:** Using LLM(large language models) can be helpful in giving human like answers , but it run into hallucination when not trained on validate data and keeping it up to date.

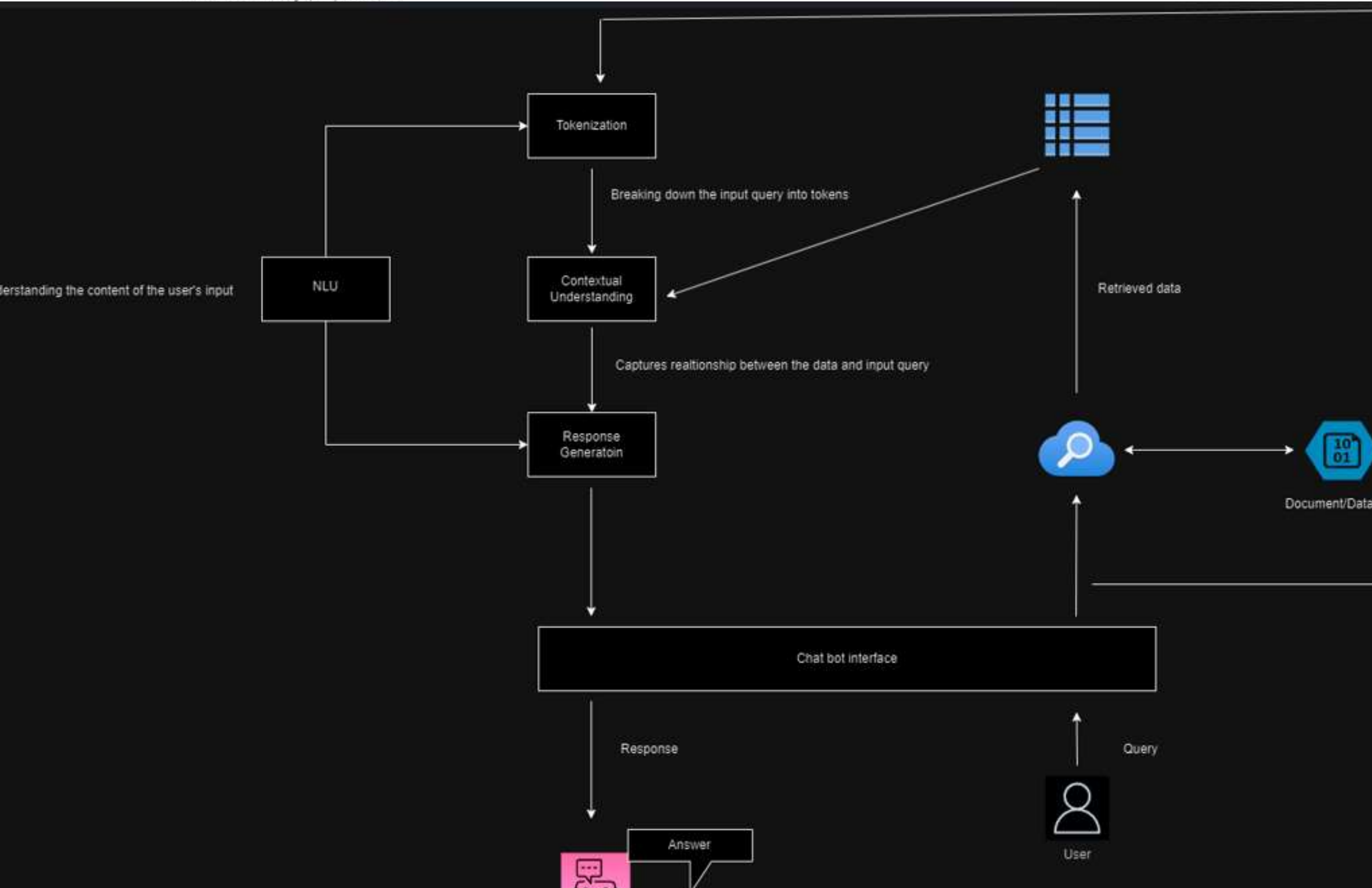
4. **Gratitude Journal**: Giving details of daily task to the chatbot is a difficult task for the users who have a busy schedule.

Objectives

The objectives of our project can be stated as:

- ❖ A chatbot which resolves the query of students at any given time, reducing the dependency/workload on human counsellor.
- ❖ Able to give the answer in simplest and easy to understandable form; mimicking humans.
- ❖ Generative that is will be able to give the answer to the query in multiple ways like tabular, different languages , points based on user prompt; also give the reference to the specific part of the documentation from which the answer has been extracted (reference).

Architecture



Modules

- Input Preprocessing and Tokenization: The initial text queries undergo preprocessing and tokenization, which breaks them down into manageable units. Able to give the answer in simplest and easy to understandable form; mimicking humans.
- Contextual Comprehension via Transformer Architecture: A GPT (Generative Pre-trained Transformer) model, like GPT-3, leverages a deep neural network based on the Transformer architecture. This architecture captures contextual nuances and relationships within the input text.
- Natural Language Understanding (NLU): The NLP module comprises a range of methods, including intent recognition, sentiment analysis, and entity recognition. These techniques are employed to comprehend and categorize student queries effectively.
- Context Management: Within the NLP module, this component is responsible for preserving a continuous conversation context, ensuring that the chatbot maintains a coherent understanding and delivers relevant responses.